

Sprint 1 Demo -- IBMS (Integrated Business Management Suite)

Field	Detail
Project Name	Integrated Business Management Suite (IBMS) v2.0
Team	Shivansh Srivastava (Product Developer), Prahallad Padhan (Product Owner), Ranveer Rai Khare (Scrum Master)
Sprint	Sprint 1
Demo Date	April 12, 2026
Duration	~15 minutes
Environment	Local (localhost:8000) -- FastAPI + MongoDB 8.2 + Vanilla JS SPA

1. Demo Objective

Demonstrate all 10 user stories delivered in Sprint 1 by walking through the running IBMS application, exercising the API endpoints, and showing the live frontend dashboard.

Sprint Goal Recap: Build a production-grade, AI-first enterprise management platform with real-time dashboards, AI-powered analytics, enterprise-grade security, and multi-environment deployment support.

2. Pre-Demo Checklist

Step	Command / Action	Expected State
1	Start MongoDB: net start MongoDB	MongoDB running on port 27017

Step	Command / Action	Expected State
2	Activate venv: .venv2\Scripts\activate	Python 3.12 venv active
3	Start server: uvicorn server:app --host 0.0.0.0 --port 8000 --reload	Server running at http://localhost:8000
4	Open browser: http://localhost:8000	IBMS login screen renders
5	Verify MongoDB: check console for MongoDB async connected	Database connected

3. Demo Script -- Feature Walkthrough

Demo 1: User Registration & Secure Login (US-01) -- 2 min

Step	Action	What to Show
1.1	Open http://localhost:8000 in browser	Auth overlay with Login/Register toggle
1.2	Click "Register" tab	Registration form appears
1.3	Type a weak password (e.g., abc)	Password strength meter shows Weak (red)
1.4	Type a strong password (e.g., Secure@2026!)	Strength meter shows Strong (green)
1.5	Fill in username, email, password -> Submit	Success toast -> auto-login -> Dashboard
1.6	API Demo: Show POST /api/auth/register response in DevTools	JWT access token + CSRF token returned
1.7	Show MongoDB users collection	Password stored as bcrypt hash (\$2b\$12\$...), not plaintext

Acceptance Criteria Validated: Password strength validation, bcrypt hashing, JWT issuance, device fingerprint capture, frontend overlay.

Demo 2: Two-Factor Authentication (US-02) -- 1.5 min

Step	Action	What to Show
2.1	Call POST /api/auth/2fa/setup with Bearer token	Returns TOTP provisioning URI + secret
2.2	Show provisioning URI is compatible with authenticator apps	URI format: otpauth://totp/IBMS:user@...
2.3	Call POST /api/auth/2fa/confirm with a valid TOTP code	2FA enabled successfully
2.4	Logout and log back in	Login form now shows the 2FA code input field
2.5	Enter TOTP code -> Login	Login succeeds with 2FA
2.6	Call POST /api/auth/2fa/disable	2FA disabled, login reverts to password-only

Acceptance Criteria Validated: TOTP setup/confirm/disable, dynamic 2FA field on login, provisioning URI generation.

Demo 3: Role-Based Dashboard Access (US-03) -- 1.5 min

Step	Action	What to Show
3.1	Call GET /api/auth/roles	Returns 5 roles with permission arrays
3.2	Login as admin user -> call GET /api/auth/me	Shows resolved permissions for admin role
3.3	Show sidebar -- all 11 pages visible	Admin sees all navigation items
3.4	Login as viewer user	Sidebar shows only Dashboard and basic pages
3.5	As viewer, call GET /api/audit/log	Returns 403 Forbidden (requires audit.view)

Step	Action	What to Show
3.6	As admin, call GET /api/audit/log	Returns paginated audit events

Acceptance Criteria Validated: 5-tier RBAC, permission inheritance, sidebar visibility, 403 on unauthorized access.

Demo 4: Real-Time KPI Dashboard (US-04) -- 2 min

Step	Action	What to Show
4.1	Navigate to Dashboard page	9 KPI cards with animated counters load
4.2	Point out the WebSocket status indicator	Green dot = connected to /ws/kpi
4.3	Wait 15 seconds	KPI values update automatically (live push)
4.4	Open DevTools -> Network -> WS tab	Show WebSocket frames arriving every 15s
4.5	Call GET /api/dashboard	JSON snapshot: revenue, margin, risk, compliance, etc.
4.6	Call GET /api/dashboard/history	Historical data points for trend charts
4.7	Navigate to Analytics page	Chart.js KPI history trend chart renders
4.8	Show 3D tilt spotlight effect on KPI cards	Hover over cards -> GPU-accelerated tilt

Acceptance Criteria Validated: Live KPI push (15s), WebSocket auto-reconnect, Chart.js charts, Redis/in-memory cache, animated counters.

Demo 5: AI-Powered Business Insights & Copilot (US-05) -- 1.5 min

Step	Action	What to Show
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Step	Action	What to Show
5.1	Call GET /api/ai/insights	4 insights: anomaly, trend, prediction, recommendation
5.2	Call GET /api/ai/anomalies	Active anomalies in transaction volume
5.3	Call POST /api/copilot/ask with {"query":"show risk details"}	AI returns risk metrics from KPI data
5.4	Call POST /api/forecast	Prophet Ensemble v2 forecast with confidence bands
5.5	Navigate to AI Insights page in frontend	AI insight cards rendered in the UI

Acceptance Criteria Validated: AI insights (4 types), anomaly detection, NL copilot, forecast with CI bands, recommendations.

Demo 6: Risk Scoring & Fraud Detection (US-06) -- 1.5 min

Step	Action	What to Show
6.1	POST /api/risk/score with {"transaction":{"amount":75000,"vendor":"new"}}	Risk score 0-100 with factor breakdown
6.2	POST /api/risk/composite with {"factors":{"amount":80,"behavior":50,"compliance":30}}	Weighted score: $(0.5 \times 80) + (0.3 \times 50) + (0.2 \times 30) = 61.0$
6.3	POST /api/fraud/detect with {"transaction":{"amount":100000,"vendor":"unknown"}}	Fraud score (0-1) + flag_for_review: true
6.4	POST /api/decision/evaluate with a high-risk input	Decision: approve / review / reject
6.5	Navigate to Risk & Fraud page in frontend	Risk scoring form + results displayed

Acceptance Criteria Validated: Transaction risk (0-100), composite weighted risk, Isolation Forest fraud, decision routing.

Demo 7: Compliance Checking & Audit Trail (US-07) -- 1 min

Step	Action	What to Show
7.1	POST /api/compliance/check with {"transaction":{"amount":600000}}	passed: false, violation: MISSING_HIGH_VALUE_APPROVAL
7.2	POST /api/compliance/check with {"transaction":{"amount":600000,"approval_ref":"APR-001"}}	passed: true, no violations
7.3	GET /api/audit/log (as admin)	Paginated audit events with timestamps
7.4	Show audit entries in MongoDB audit_logs collection	Events with base64-encoded details

Acceptance Criteria Validated: Control-set validation, violation codes, audit trail persistence, RBAC-gated audit log.

Demo 8: Budget Optimization & Dynamic Pricing (US-08) -- 1 min

Step	Action	What to Show
8.1	POST /api/budget/optimize with {"items":[{"name":"Marketing","amount":50000}, {"name":"R&D","amount":80000}], "growth_target":15}	Optimized totals with growth multiplier
8.2	POST /api/pricing/suggest with {"demand_index":0.8, "stock_level":200, "competitor_price":999}	Recommended dynamic price

Step	Action	What to Show
8.3	Navigate to Budget and Pricing pages in frontend	Forms + result output displayed

Acceptance Criteria Validated: Budget optimizer with growth target, dynamic pricing with demand/stock/competitor weights.

Demo 9: Lead Scoring & Inventory Prediction (US-09) -- 1 min

Step	Action	What to Show
9.1	POST /api/leads/score with {"engagement":{"email_opens":15,"page_visits":30,"demo_requests":2},"fit":{"company_size":"enterprise","industry":"fintech","budget":500000}}	Score + tier (hot/warm/cold)
9.2	POST /api/inventory/predict with {"product_id":"SKU-001","lead_time_days":14}	Reorder point prediction
9.3	Navigate to Lead Scoring page in frontend	Lead scoring form + tier output

Acceptance Criteria Validated: Lead qualification (55% engagement + 45% fit), tier classification, inventory reorder prediction.

Demo 10: System Health Monitoring & Deployment (US-10) -- 2 min

Step	Action	What to Show
10.1	GET /api/health	status: healthy, uptime, request count, error rate

Step	Action	What to Show
10.2	GET /api/metrics	System metrics snapshot
10.3	GET /api/system/status	Top endpoints, cache state, detailed status
10.4	Show response headers in DevTools	X-Trace-Id, X-Response-Time-Ms, security headers
10.5	Navigate to System Health page in frontend	Server status dashboard
10.6	Show docker-compose.yml	Nginx + Web + Redis + MongoDB orchestration
10.7	Show deploy/k8s/ibms-core-deployment.yaml	3 replicas, resource limits, health probes
10.8	Show deploy/aws/terraform/main.tf	VPC, subnets, ElastiCache, security groups

Acceptance Criteria Validated: Health/metrics/status endpoints, trace headers, security headers, Docker/K8s/Terraform configs.

4. Live Database Verification

Collection	Documents	Key Data
users	3	admin, analyst, shiv1 -- bcrypt-hashed passwords
kpi_snapshots	24	Revenue, margin, risk, compliance -- every 15s
kpi_latest	1	Materialized view -- latest KPI for instant reads
audit_logs	2+	Login, register events -- timestamped, base64
refresh_tokens	1+	Active refresh tokens with TTL expiry index
csrf_tokens	1+	CSRF tokens with 1-hour TTL expiry index
notifications	1+	System notifications
alerts	0	Ready -- indexed by severity + resolved status

Collection	Documents	Key Data
ai_recommendations	0	Ready -- indexed by company + recommendation type
enterprise_profiles	0	Ready -- unique index on company_id
webhook_logs	0	Ready -- indexed by processed status
decision_rules	0	Ready -- indexed by module + enabled status
rate_limits	0+	IP-based rate limit tracking

Total Collections: 13 with production-ready indexes and TTL policies.

5. Demo Summary -- Sprint 1 Delivered Features

#	Feature Area	Endpoints Demonstrated	Story Points	Verdict
1	Registration & Login	5 endpoints	8	[x] Fully Working
2	Two-Factor Authentication	3 endpoints	5	[x] Fully Working
3	Role-Based Access Control	3 endpoints	8	[x] Fully Working
4	Real-Time KPI Dashboard	3 endpoints + WS	8	[x] Fully Working
5	AI Insights & Copilot	4 endpoints	8	[x] Fully Working
6	Risk Scoring & Fraud Detection	4 endpoints	8	[x] Fully Working
7	Compliance & Audit Trail	2 endpoints	5	[x] Fully Working
8	Budget & Dynamic Pricing	2 endpoints	5	[x] Fully Working
9	Lead Scoring & Inventory	2 endpoints	5	[x] Fully Working
10	System Health & Deployment	3 endpoints + configs	8	[x] Fully Working
	Total	31+ endpoints	68 pts	10/10 Done

6. Q&A / Stakeholder Feedback

Questions anticipated:

Question	Answer Summary
How is data persisted?	MongoDB 8.2 (13 collections, TTL indexes, materialized views)
What if Redis goes down?	Graceful fallback to in-memory dict -- zero downtime
How are tokens secured?	JWT HS256 (30-min TTL) + refresh rotation (7-day) + device binding
What ML model is used for fraud?	Isolation Forest (scikit-learn) -- unsupervised, no labeled data needed
Can this scale horizontally?	K8s 3-replica deployment ready; stateless backend allows scaling
What's the test coverage?	14 tests (5 unit + 9 integration), all passing -- improving Sprint 2

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